



Academic Project:

Development of a Restaurant Management Web Application

Framework: Django (Python)

DBMS: MySQL

Python Programming Course ----- Academic year 2025-2026

Lecturer: Mbietieu Amos Mb., ENSPY

Yaounde

1. CONTEXT AND JUSTIFICATION OF THE PROJECT

In an economic environment marked by the accelerated digitization of services, the restaurant sector is a strategic area for technological innovation.

A modern restaurant can no longer limit itself to manual management of recipes, orders, and inventory. It must:

- Optimize its costs;
- Automate its processes;
- Ensure the traceability of operations;
- Improve the customer experience;
- Have real-time financial indicators.

The analytical work carried out has enabled the design of a comprehensive MySQL database covering:

- Revenue management;
- Order management;
- Employee management;
- Inventory management;
- Delivery management;
- Finance management.

The next step is to take this database to the next level by transforming it into a professional web application developed in Python with Django.

2. PROJECT OBJECTIVES

2.1 General Objective

Develop a comprehensive restaurant management web application based on Django and connected to an existing MySQL database.

2.2 Specific Objectives

Students will be required to:

- Implement Django models corresponding to the existing SQL schema;
- Configure MySQL as the main DBMS;
- Develop a customized administration interface;
- Design user-friendly interfaces;
- Implement business rules;
- Generate dynamic financial reports;
- Ensure data security.

3. FINANCIAL AND INTELLECTUAL INTEREST OF THE PROJECT

3.1 Financial Interest

Such an application would enable:

- Reduce losses due to human error;
- Improve inventory control;
- Optimization of supplies;
- Analysis of commercial performance;
- Increased profitability.

Students will gain a practical understanding of how an information system impacts a company's economic performance.

3.2 Intellectual interest

This project involves:

- Advanced relational modeling and UML language;
- Django ORM(Object-Relational Mapping);
- MVC architecture (MVT in Django);
- Web security;
- Authentication and role management;

- Statistics generation;
- Deployment.

It forms a genuine bridge between academic theory and industrial reality.

4. PROJECT ORGANIZATION

The following organization is required to carry out this project:

- Work in groups of 4 students;
- Agile methodology recommended;
- Mandatory use of Github;
- Mandatory technical documentation.

5. FUNCTIONAL SCOPE

The application must cover the modules listed in the following table:

5.1 User Management	5.2 Revenue Management	5.3 Product Management
Types of users: a) Administrator; b) Director; c) Head chef; d) Cook; e) Server; f) Stock manager; Features: 1) Secure authentication; 2) Role management; 3) Specific permissions	a) Creation/modification/deletion; b) Recipe-ingredient association; c) Required quantities; d) Cooking time; e) Chief in charge.	a) Dish catalog; b) Categorization; c) Dynamic pricing



--	--	--

Here are the other features:

5.4 Order Management	5.5 Inventory Management	5.6 HR Management	5.7 Dashboard
a) On-site orders; b) Delivery orders; c) Automatic calculation of the amount; d) Invoice generation.	a) Automatic update after order; b) Critical threshold alerts; c) History of variations.	a) Employee files; b) Assignments; c) Payroll calculation.	a) Monthly turnover; b) Number of orders; c) Best-selling products; d) Supply expenses.

6. TECHNICAL CONSTRAINTS

The software engineering workshop primarily consists of the following elements:

- Framework: Django;
- Language: Python 3.x;
- DBMS: MySQL (mandatory);
- Django ORM required;
- MVT architecture respected;
- Structured and commented code.

7. NON-FUNCTIONAL REQUIREMENTS

Some non-functional requirements:

- Ergonomic interface;
- CSRF security enabled;
- Protection against SQL injections;
- Optimized performance;
- Logging of critical actions.

8. EXPECTED DELIVERABLES

Each group must provide the following via the GitHub link:

1. Complete source code;
2. Configured MySQL database;
3. Technical report (25-35 pages);
4. User manual;
5. Oral presentation with demonstration.

9. EVALUATION CRITERIA

Here are some criteria for evaluating your project:

1. Quality of architecture;
2. Compliance with the relational schema;
3. Implemented features;
4. Code quality;
5. Security;
6. Innovation;
7. Clarity of presentation.

10. STRATEGIC CONCLUSION

This project is not just an academic exercise. It represents a real immersion in:

- Software engineering;
- Information systems engineering;
- The economic impact of digital solutions.

Those who take on this challenge will not only build an application. They will build a professional skill.